VIDHATA+

Plastics India Pvt. Ltd.

precision

tooling

quality

we Care as we Deliver...

Engineering Life
with Care.

At VIDHATA, our pursuit of engineering excellence is a reflection of our commitment to matters closest to our customers' growth—such as quality, precision, and operational trust.

While our toolroom and injection moulding technologies are truly **state-of-the-art**, our corporate execution model is **state-of-the-heart**.

Innovations, zero-defect quality standards, and end-to-end tooling options flow forth from our passion to serve medical, electronic, and industrial markets in a meaningful way.

Directory & Index

This index details the complete directory of engineering capabilities, manufacturing infrastructure, cleanroom medical lines, and quality verification systems available under one roof at Vidhata Plastics India Pvt. Ltd.

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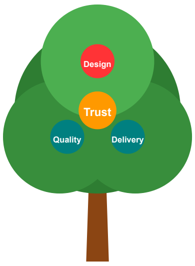
Our Vision

To become a globally trusted manufacturing partner for medical, electronic, and industrial sectors.



Our Mission

Deliver world-class plastics engineering through innovation, quality, and customer-focused execution.

Precision Plastics at work...

Company Profile

Vidhata Plastics India Pvt. Ltd. was established to provide high-quality engineering plastic components at a very competitive cost. We match Chinese pricing structures while delivering superior quality and technical support.

Experienced Engineers

Our core asset is our highly trained design and manufacturing engineers. We apply continuous training in Six Sigma guidelines to eliminate molding defects, optimize cycle times, and maintain dimensional tolerances down to ±5 microns.

Production Facilities

Our infrastructure includes over 400,000 sq ft of manufacturing area, with dedicated Class 100,000 and Class 10,000 cleanroom moulding zones. High-speed Haitian machines, CNC toolrooms, and automated material systems support our operations.

we Care as we Deliver...

Quality Assurance

Vidhata is committed to maintaining a strict Quality Management System (QMS) compliant with ISO 9001:2015 and ISO 13485 (Medical Quality Standard). We practice a PREVENTIVE APPROACH to manufacturing, enforcing raw material verification, moisture control, and inline parameter checks.

Continuous Improvement

Our dedicated engineers focus on continuous cycle-time reductions and process automation. We utilize mold flow simulations early in the design stage to eliminate parting lines, gate flaws, and mechanical weaknesses before building the hard tools.

VIDHATA+

PRECISION PLASTICS

Q. M. S.

01. Product Design & CAD Modeling

Series: 101-110 Tool Design & Engineering

Our in-house design center uses advanced CAD platforms for 3D part design, reverse engineering, and prototyping. By creating highly precise digital mockups, we prevent mechanical design errors early in the cycle.

DESIGN CENTER HIGHLIGHTS

- **3D CAD Modeling:** Drafting complex solid models with drafts, wall offsets, and optimized parting lines.
- **DFM (Design for Manufacturing):** Optimization of draft angles, rib thickness, and boss dimensions.
- **Mold Flow Simulation:** Gate location positioning, filling analysis, cooling mapping, and shrinkage prediction.
- **Rapid Prototyping:** Visualizing physical parts before making the hard tooling.

101 3D CAD Design

102 DFM Optimization

103 Mold Flow Simulation

02. High-Precision Injection Molds

Series: 111-120 Hard Tool Mould Engineering

Vidhata designs and manufactures custom injection moulds in-house. We work with premium mold steel (H13, S136, NAK80) heat-treated for long-term wear resistance, guaranteeing lifecycles of over 1 Million shots.

MOULD BASE	CAVITY STEEL	RUNNER TYPE	LIFESPAN
LKM, DME	S136 Hardened	Yudo Hot Runner	1,000,000+ shots
LKM, Hasco	H13 Hardened	Cold / Semi-Hot	500,000+ shots
Standard C50	P20 Pre-hardened	Cold Runner	300,000+ shots

TOOL ROOM SPECIALTIES

- **Multi-Cavity Molds:** High-cavitation (up to 64 cavities) for high-speed medical disposables moulding.
- **Hot Runner Systems:** Valve-gated hot runner systems for clean, sprue-less part gating.
- **Sliding Cores & Lifters:** Intricate slide mechanisms for undercuts and side ports.

03. Extrusion Dies & Polyurethane Tooling

Series: 121-130 Extrusion & PU Casting Molds

We design and manufacture specialized tooling including high-flow profile extrusion dies, downstream calibrators, and cast polyurethane moulds used for industrial footwear and shock-absorbing parts.

SPECIALIZED TOOLING CAPABILITIES

- **Extrusion dies:** Flow-optimized head designs for polycarbonate and PVC profile extrusion.
- **Vacuum Calibrators:** Water-jacketed calibrator units ensuring highly precise outer tube diameters.
- **PU casting moulds:** Multi-station rotating table moulds compatible with DESMA and BASF systems.
- **Prototype Molds:** Soft tool aluminum moulds for low-volume casting trials.

121 Profile Dies

122 Water Calibrators

123 PU Sole Moulds

04. Precision Machining Centers

Series: 131-140 In-House Machining Infrastructure

Our high-end CNC and EDM machinery room runs 24/7. Equipped with Metrology and inspection interfaces, we process hardened core pins, stripper rings, and complex slide assemblies with high reliability.

MACHINE TYPE	MAKE & SPEC	AXIS & TRAVELS	TOLERANCES
CNC Machining	DMG Mori CNC Mill	3-Axis, 12,000 RPM	±5 µm
High-Speed CNC	Makino CNC Center	3-Axis, 20,000 RPM	±2 µm
Wire EDM	Fanuc Wire Cut	4-Axis Spark wire	±3 µm
Die Sink EDM	Charmilles EDM	Z-axis CNC sparker	±5 µm

MACHINING ACCURACIES

- **CNC Milling:** DMG Mori spindles operating at 20,000 RPM, preventing heat-distortion in mold cavities.
- **Wire EDM:** Fanuc wire discharge machines providing slot cuts with tolerances down to ±2 microns.
- **Surface Finish:** Precision grinding matching mirror polish requirements.

05. Cleanroom Injection Moulding

Series: 201-210 ISO Class 8 Sterile Injection Lines

We operate dedicated, climate-controlled ISO Class 8 (Class 100,000) cleanrooms for medical components. Particulate counts, humidity, and airflow are monitored 24/7 to ensure zero-defect sterile moulding.

CLEANROOM MOULDING CAPABILITIES

- **Sterile Moulding:** Low-particulate injection zones dedicated to fluid-pathways.
- **Raw Materials:** Medical-grade polycarbonate (PC), polypropylene (PP), and cyclic olefins (COC).
- **Material Conveying:** Closed-loop vacuum lines convey resins directly from external dryers to machine hoppers.
- **Automation:** Cleanroom-rated pick-and-place robots strip parts and deposit them directly onto sterile conveyors.

201 Cleanroom Class 8

202 USP Class VI Resins

203 Robot Demolding

06. High-Volume Injection Moulding

Series: 211-220 Heavy Clamping Force Machinery

Our primary manufacturing bays run a fleet of Haitian injection moulding machines. With clamping capacities from 120T to 250T, we support continuous, high-volume production of electronic charger cases, POS meter housings, and industrial battery containers.

MACHINE SIZE	CLAMPING FORCE	PLATEN DIMENSIONS	RESIN OPTIONS
Haitian 120T	1200 kN	410 x 410 mm	PP, ABS, PMMA, PC
Haitian 160T	1600 kN	470 x 470 mm	PC, Polyacetal, ABS
Haitian 200T	2000 kN	530 x 530 mm	PA66, Glass-filled PBT
Haitian 250T	2500 kN	610 x 610 mm	Heavy Battery ABS/PP

HIGH VOLUME INFRASTRUCTURE

- **Automated Conveying:** Centralized material conveying prevents mix-ups and dust issues.
- **Chiller Systems:** Water cooling circuits maintain stable mold temperatures, reducing cycle time.
- **Annealing:** In-house stress-relieving ovens normalize polycarbonate parts to prevent crack lines.

07. Injection Blow Moulding (IBM)

Series: 221-230 Sterile Closures & Containers

We operate a 60T dedicated Injection Blow Moulding line inside our cleanrooms. This process combines precise neck finish injection with stress-free container blow moulding, ideal for pharmaceutical droppers and bottles.

IBM PRODUCT CAPABILITIES

- **Dropper Bottles:** 5ml to 100ml ophthalmic eye dropper bottles with zero leakage risk.
- **Flip-Off Caps:** High-precision caps, flip-off seals, droppers, and blood collection caps.
- **Child Resistant Closures (CRC):** Custom design squeeze-and-turn closures for medical syrups.
- **Material Grades:** Certified food-contact and medical grades (LDPE, HDPE, PP).

221 Injection Blow

222 LDPE Dropper Bottles

223 Leak-Proof Neck Finish

08. Polycarbonate & Profile Extrusion

Series: 231-240 Precision Tube & Profile Extrusion

We operate state-of-the-art extrusion lines with online diameter calibrators. We specialize in high-optical-grade polycarbonate extrusion for diffuse lighting and strict medical-grade tubing profiles.

PROFILE TYPE	MATERIALS	WALL THICKNESS TOLERANCE	APPLICATION
LED Diffuser Profiles	Optical Polycarbonate	±0.05 mm	Philips, Orient Lighting
Medical Tubing	Medical PVC / TPU	±0.02 mm	Catheters, IV lines
Light Covers	UV-Stabilized PMMA	±0.08 mm	Outdoor Street Lights

EXTRUSION FEATURES

- **PC Diffusers:** Frosted and micro-prism co-extrusion diffusers for LED light fixtures.
- **Batten Housings:** Extruded batten housings, street light polycarbonate covers.
- **Medical Tubing:** Cleanroom extrusion of multi-lumen PVC and polyurethane catheter tubes.

09. Polyurethane (PU) Casting & Processing

Series: 241-250 DESMA Multi-Station PU Casting

Vidhata integrates advanced polyurethane processing using BASF Elastogran chemical systems and DESMA rotary casting machine tables. We output highly durable components matching strict wear requirements.

PU PROCESSING HIGHLIGHTS

- **DESMA Casting:** 24-station rotating mold carrier lines ensuring high throughput.
- **BASF Chemistry:** Utilizing premium Elastogran polyols and isocyanates for density control.
- **PU Soles production:** Over 3 Million pairs manufactured annually for Ecco, Adidas, and Bata.
- **Industrial Dampeners:** Custom polyurethane gaskets, bushes, and high-impact industrial pads.

241 DESMA Systems

242 BASF Elastogran

243 3M+ Pairs Soles

10. Medical Disposable Device Assembly

Series: 301-310 Closed-System Cleanroom Assembly

We assemble multi-part medical disposables inside Class 8 cleanrooms. From dialysis housing caps and ports to I.V. cannula components and catheter manifolds, we maintain sterile joint integrity and traceablity.

MEDICAL ASSEMBLY PROTOCOLS

- **Cleanroom Bonding:** Controlled-solvent and UV-curing adhesive lines.
- **Tube Insertion:** Semi-automated solvent insertion of medical tubes into manifolds.
- **Device Traceability:** Maintains complete Device Master Records (DMR) for lot traceablity.

DEVICE CATEGORY	JOINING METHODS	TEST COMPLIANCE
Dialysis End Caps/Ports	Ultrasonic welding, solvent bonding	ISO 8637 compliant leak tests
Catheter Manifolds	UV-glue bonding, solvent bonding	Pressure decay leak checking
I.V. Cannula Parts	Press fit, adhesive bonding	ISO 10555 standard tests

11. Component Assembly & Secondary Services

Series: 311-320 Finished Goods Mechanical Assembly

Beyond injection moulding, Vidhata provides extensive post-moulding capabilities. We assemble multi-part electronics housings, insert metal screws/gaskets, and perform secondary visual and structural tests.

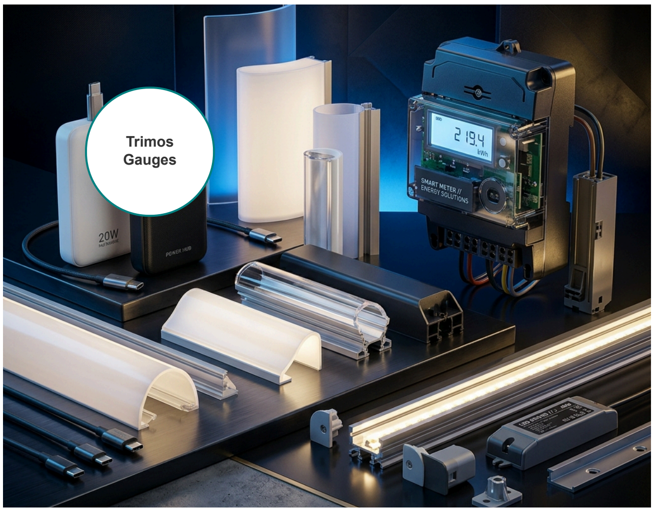
ASSEMBLY & SECONDARY CAPABILITIES

- **Ultrasonic Welding:** High-frequency sound waves fuse plastics without solvent residue.
- **Pad Printing & Laser Marking:** Precise marking of logo, model, ratings, and serial lot codes.
- **Threaded Inserts:** Heat and ultrasonic insertion of brass threaded nuts into ABS housings.
- **Gasket Fitting:** Automated silicon gasket dispensing for IP67 water-tight POS/charger housings.

311 Ultrasonic Weld

312 Pad Print Codes

313 IP67 Test line



METROLOGY LAB ASSETS

- **Trimos Gauges:** Digital 1D/2D vertical height gauges for step, diameter, and slot check.
- **Mitutoyo Projector:** Profile projectors check angles, drafts, and radii.
- **MFI testing:** Melt Flow Index checks raw resin viscosity to verify batch properties.

12. Metrology Laboratory & Calibration

Series: 401-410 Dimensional Quality Verification

Our in-house Metrology lab acts as our quality control checkpoint. We verify First Article Inspection (FAI) dimensional compliance against drawings and maintain records for ISO audits.

INSTRUMENT CATEGORY	MAKE & SPEC	DIMENSION VERIFIED	RESOLUTION
Height Gauge	Trimos V5 Digital	Step height, outer diameters	0.001 mm
Profile Projector	Mitutoyo PJ-A3000	Draft angles, small radius	0.001 mm
Moisture Analyzer	Sartorius MA35	Raw resin moisture content	0.01 %
Viscosity Index	Melt Flow Indexer (MFI)	Resin melt flow rate batch index	0.01 g/10min

13. Functional Leak, Flow & Pressure Testing

Series: 411-420 Functional Verification Equipment

For medical devices and water-tight electronic housings, we conduct 100% inline functional tests. We use automatic pressure decay check systems to verify joint seal integrity under strict ISO directives.

FUNCTIONAL TESTING CAPABILITIES

- **Pressure Decay Leak Test:** Ateq testers check micro-leakage down to 0.1 Pa/sec.
- **High Pressure Testing:** Verifying IV catheter manifolds and dialysis lines under pressure.
- **Flow Rate verification:** Flow rate checking under gravity water heads (Milli-bar range).
- **Destructive Burst Test:** Pneumatic burst testers measure weld joint shear thresholds.

411 Pressure Decay

412 Flow Verification

413 Lot Burst Test

14. Sterile Blister Packaging & Sterilization

Series: 501-510 Sterile Packaging & Logistics

Our cleanrooms house thermoforming blister and Tyvek pouch heat-sealing machines. We establish sterile barriers for medical disposables under ISO 11607 directives, managing sterilization coordination.

PACKAGING TYPE	SUBSTRATE MATERIAL	STERILITY INDICATOR	STANDARD COMPLIANCE
Tyvek Pouches	DuPont Tyvek 1073B / Film	Chemical process dot indicator	ISO 11607 sterile seal
Blister Trays	Medical grade PETG / Tyvek lid	EO / Gamma color tag	EN 868 medical pack
Film Pouches	PE/PA laminated film	Chemical process dot indicator	ISO 11607 compliant

STERILE BARRIER FEATURES

- **Tyvek Sealing:** Medical-grade heat sealers check seal peeling strength parameters.
- **Blister Packaging:** Automated thermoforming blister packing machines create trays.
- **ETO Coordination:** Manage EO (Ethylene Oxide) sterilization cycles with certified labs.

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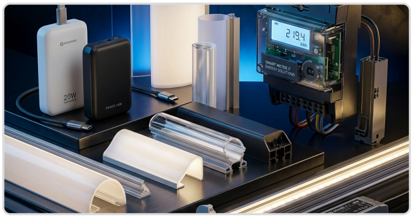
Plastics India Pvt. Ltd. — Precision Tooling & Moulding





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